

insights needed to boost yield, cut scrap, and schedule maintenance proactively.

Predictive modelling. Build models that can analyse data to forecast outcomes rather than just reporting historical performance.

Resource optimisation. Use forecasts to create precise schedules. Instead of firefighting breakdowns with whoever is free and available for parts, the right technician is scheduled, correct spare parts are staged at the work cell, and shift mixes are aligned with predicted demand—eliminating scramble and keeping costs under control.

Continuous refinement. Establish feedback loops to compare forecasted outcomes with actual results, tweak model parameters, and feed in the newest data. Continuous refinement improves accuracy over time.

Organisations that have implemented this framework for over a decade report average cost reductions of 15-25% in first-year operations, primarily through reduced warranty claims, optimised maintenance scheduling, and improved production yield.

How piling up data drains profits

A massive 20-year dataset from a flight carrier, containing 2.5 million flight hours of aircraft performance data, shows electronics components face dramatically different operating conditions across markets.

Working with enterprises from Apple's supply chain to aerospace

It does not matter if an enterprise is small or has little to no historical data; it is time for SMEs and medium-scale enterprises also to adopt this technology, learning from the successes of large enterprises.

Do not reinvent the wheel. Take baby steps: start with a pilot on one line, then scale systematically. Organisations can grow rapidly and exponentially while always anchoring back to a clear business objective, rather than chasing shortcuts.

By analysing datasets, we found electronics components face dramatically different operating conditions across markets, creating unique challenges for manufacturers. Products designed and tested in controlled Japanese or German facilities often fail when deployed in extreme climates, from Srinagar's cold to Chennai's heat. Sensors that work perfectly in clean environments experience significantly higher failure rates in Rajasthan's dusty conditions or Jharkhand's coal-mining regions.

This geographic complexity means the same product requires different component specifications for Russia, India, and Africa, a reality that only integrated analytics can address systematically.

In India's context, this scale becomes critical: Bombay High alone produces 30% of the country's crude oil requirements, generating enormous data streams that traditional analysis methods cannot handle effectively.

Highlighting these local conditions convinces leaders that integrated analytics is not a luxury; it is a necessity for reliability and customer trust.

systems reveals a consistent crisis: organisations possess vast analytical potential trapped in fragmented systems. This fragmentation costs the electronics industry billions annually due to component failures in electric vehicles, with warranty claims averaging \$2000-\$5000 per incident.

Immediate action for supply chain leaders

The first step with this piled-up data is to audit all datasets and make inventory of all systems generating operational data. Most organisations discover 40-60% more usable data than they initially estimated.

Calculate poor quality costs. Quantify current expenses from scrap, rework, and warranty claims. This typically represents 8-12% of revenue for electronics manufacturers.

Identify integration opportunities. Start with production and quality data integration, as it is usually the fastest path to measurable ROI.

Analytics should go beyond traditional metrics

Electronics manufacturing has long relied on statistical process control methods—X-bar and R-bar charts, control limits, and variance analysis. These approaches work for stable and controlled environments, but usually break down under modern complexities.

Consider automotive electronics suppliers serving diverse markets. Components performing flawlessly in German testing facilities may behave differently when deployed in extreme climates, from Kashmir's cold to Chennai's heat. Traditional quality control cannot account for such environmental variables, leading to field failures that integrated analytics could predict and prevent.

How a company should adapt to change

The most significant change is organisational, not technological. Chief Data Officers did not exist a decade ago, but now occupy board-level positions because analytics directly impacts key business metrics, production yield, quality enhancements, and warranty cost reductions.

A myopic vision is insufficient. A limited data set or isolated efforts will fail. The IoT program must evolve over time, bringing in design engineering, supply chain, and product management teams. Integration and a long-term vision are critical for success. **EFY**

This article is based on the session titled Advanced Analytical Innovation for Electronics IoT Program Success Using AI-ML and Event Stream Processing, delivered by Arvind Kumar Prasad, Founder & Director of Vortexa Technology Advisors, at IEW25 in Bengaluru. The article was transcribed and curated by Janarthana Krishna Venkatesan, journalist at EFY.